Simeon **Tran**

simeonat.github.io | SimeonAT | simeon-tran

Education

University of California, Irvine — *Irvine, California*

PH.D. IN NETWORKED SYSTEMS — *Jun. 2026 - PRESENT*

Research Topic — Secure Situational Awareness in Drone Swarms

Advisor: Prof. Gene Tsudik

University of California, Santa Cruz — *Santa Cruz, California*

M.S. IN COMPUTER SCIENCE AND ENGINEERING — *Sept. 2023 - Dec. 2025*

B.S. IN COMPUTER SCIENCE — *Jul. 2019 - Jun. 2023*

Publications

Comparative Study of Encryption Algorithms in Battery Powered Thread© Networks for Smart Homes — *Jan. 2024 - Dec. 2025*

HTTPS://SIMEONAT.GITHUB.IO/RESEARCH/THESIS

Poster Abstract Published in SenSys 2026

- Modified OpenThread networking stack by replacing AES-CCM with the ASCON lightweight encryption scheme
- Extended ESP-IDF to enable 802.15.4 ESP32 devices to run custom OpenThread implementation supporting ASCON
- Evaluated network performance and power consumption to validate cryptographic changes under real hardware constraints in simulated smart home environment
- Results show no significant impact on battery lifetime or network performance when ASCON is used instead of AES-CCM

Experience

Security & Privacy Research Outfit (SPROUT) at UCI — *Sep. 2026 - PRESENT*

GRADUATE STUDENT RESEARCHER

- Migrated a UAV cybersecurity simulation framework from ArduPilot to PX4 to evaluate secure a situational awareness protocol for drone swarms
- Root-caused simulation failures when running multiple vehicles, scaling framework to run 20 concurrent drones
- Setup Jetson Orin NX and documented hardware configurations for two Pixhawk drones in preparation for live swarm testing

Sphinox Inc. — *Jun. 2022 - Aug. 2026*

FULL STACK WEB DEVELOPER & NETWORK ADMINISTRATOR

- Owned development and long-term operation of Next.js CRM for barber shop customer rewards program containing more than 6000 user records
- Maintained company-wide networking infrastructure containing 5 servers and 20+ devices using VLANs, Tailscale ACLs, and administering UniFi networking equipment, securing internal services and production systems
- Built observability into server infrastructure using Prometheus and Grafana for 24/7 monitoring of uptime, performance, and failures

Tech4Good Lab at UCSC — *Jun. 2023 - Dec. 2023*

LEAD RESEARCH ASSISTANT — *Santa Cruz, California*

- Led team of 4 other students prototype to develop Express.js and OpenAI API backend service integrating ChatGPT to generate personalized learning curricula
- Supported team members new to backend development by assisting in defining backend interface using OpenAPI specification and taught them how to write unit tests using Jasmine testing framework
- Designed and conducted 3 user interviews to evaluate system effectiveness and inform iteration on backend and UX design

WEB DEVELOPER TEAM LEAD — *Apr. 2023 - Jun. 2023*

- Built cross-platform mobile deployment infrastructure using Ionic to enable preexisting Angular web applications to run on iOS and Android
- Integrated Google OAuth authentication in mobile applications using Capacitor Ionic plugin
- Wrote onboarding instructions and 44-page documentation to help new members to get acquainted with Ionic

Technical Skills

Embedded Systems	ESP-IDF, OpenThread, nRF Connect, Analog Discovery 3
Networks	MAVLink, PX4, Gazebo, Wireshark, Prometheus, Grafana, UniFi
Web Frameworks	Node.js, Deno, Next.js, Astro
Front-end	React, Angular, Vue.js
Databases	PostgreSQL, Firebase
Back-end	OpenAPI, GraphQL, Express.js
CI/CD	Docker, GitHub Actions
Programming	C/C++, Python, Typescript/Javascript, Haskell

Projects

Daily Time Manager Web Application

Sept. 2022 - Dec. 2022

https://github.com/SimeonAT/Bagel.io/tree/main

- Worked with four fellow software engineering classmates to create a web app that generates visualizations based on completed tasks on a user's to-do list
- Designed the UI for the task cards and to-do list
- Developed both the React front-end and Node.js back-end implementations for the login and register pages

Fractal Terrain Generation Program for Python Course

Apr. 2020

https://users.soe.ucsc.edu/~pang/30/s20/prog1/terrain_Simeon.txt

- Used Python Turtle Module and starter code provided by the class TA to create program that recursively generates random terrain
- Developed extra features beyond what was asked, which encouraged graders and course faculty to label this program as a "star solution" on course website (`https://users.soe.ucsc.edu/~pang/30/s20/`)

Video Game Demo using Python and PyGame

Dec. 2020 - Dec. 2021

https://github.com/SimeonAT/SpaceViewerPyGame

- Created single-player game demo where players interact with objects in outer space
- Implemented RPG-style text dialogue using object-oriented programming
- Applied knowledge of data structures by creating a binary tree to store RPG text-box objects
- Integrated pixel art sprites with Python code to create lively animations

Teaching

Baskin Engineering at UCSC

Jun. 2024 - Jul. 2024

Full Stack Web Development Teaching Assistant — *Remote*

- Held daily office hours where I helped students with debugging
- Taught students web development using React, Express, and PostgreSQL
- Helped students develop the habit of Test Driven Development (TDD)

UC Santa Cruz Learning Support Services

Sep. 2020 - Jun. 2021

Python Course Tutor — *Remote*

- Held weekly remote tutoring sessions with diverse students for Introduction to Python course
- Assisted students on Python concepts and basic object-oriented programming to help them perform well on assignments and exams
- Demonstrated knowledge of Python by creating programming examples and activities

UC Santa Cruz Academic Excellence Program

Oct. 2021 - Dec. 2021

Precalculus Tutor — *Remote*

- Was a tutor/mentor for Professor Morales-Almazan's Precalculus class, under the recommendation of the professor himself
- Helped students with Precalculus topics that they struggled with, allowing them to be better prepared for the homework and exams
- Gave freshman students from different backgrounds advice on how to navigate the university, in addition to assisting them with math